

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                 | Marks available |     |     |       |       |      |
|----------|-----|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                 | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (e) | (i)  |  | Hess' cycle drawn (1)<br><br>$\Delta_f H^\ominus - 1371 = 2(-394) + 3(-286)$ (1)<br><br>$\Delta_f H^\ominus = -275$ (1)<br><br>award (3) for cao<br>ecf possible                                                                                                                                                                                |                 | 3   |     | 3     | 2     |      |
|          |     | (ii) |  | Amir is correct – no credit<br><br>award credit only for calculations<br><br>C                    394/12<br>H <sub>2</sub> 286/2.02<br>C <sub>2</sub> H <sub>5</sub> OH          1371/46                    (1)<br><br>C                    32.8<br>H <sub>2</sub> 142<br>C <sub>2</sub> H <sub>5</sub> OH          29.8                    (1) |                 |     | 2   | 2     | 1     |      |
|          |     |      |  | Question 7 total                                                                                                                                                                                                                                                                                                                                | 1               | 8   | 5   | 14    | 3     | 6    |